

Hierarchical Multi-Stream Learning for Eye Movement-Based Schizophrenia Recognition: From Handcrafted Features to Bidirectional Graph-Transformer Ensembles

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Abstract—Schizophrenia recognition from free-viewing eye movements lacks integration between clinical biomarkers and sequence learning. We propose a Hierarchical 5-Tier Framework using the EMS dataset (208 subjects). Our pipeline scales from preprocessing (Tier 1) and feature engineering of trial-level and category-wise contrastive delta features (Tier 2) to tabular baselines (Tier 3), parallel deep sequential architectures (Tier 4), and meta-learner ensembling (Tier 5). In Tier 4, we develop GNN-CEFAM, a Graph Attention Network with projected ResNet50 visual nodes, and BiCA-HS, a Transformer-based sequence encoder. Both architectures leverage bidirectional cross-attention where clinical biomarkers and deep trajectories mutually guide each other, yielding visual attention mappings. Both architectures outperform the state-of-the-art MSNet baseline (AUC = 0.8854, $p < 0.05$). BiCA-HS achieves the strongest validated out-of-fold performance with an AUC of 0.9454, while the Tier 5 ensemble exploratory meta-learner further combines tabular and deep streams for diagnostic consensus.

Index Terms—Schizophrenia, eye movement, scanpath, graph attention network, transformer, cross-attention, ensemble learning.

I. INTRODUCTION

Schizophrenia is a neuropsychiatric disorder characterized by cognitive and social functioning deficits. Clinical diagnosis remains qualitative and prone to subjective clinician bias, motivating the search for objective biomarkers. Atypical eye movements during free-viewing paradigms represent a promising physiological biomarker. Although patients show restricted visual exploration and social cue avoidance, existing classifiers rely on static features that discard temporal sequences, and deep models like MSNet [1] rely on static density maps that ignore temporal saccade trajectories and lack fixation-level clinical interpretability. To address this, we propose a Hierarchical 5-Tier Framework. It integrates raw spatiotemporal filtering (Tier 1), feature engineering of category-wise contrastive delta features (Tier 2), and tabular baselines (Tier 3), with parallel deep architectures (Tier 4): a Graph Attention Network (GNN-CEFAM) with projected ResNet50 visual nodes, and a Transformer-based sequence encoder (BiCA-HS). Both architectures utilize a cross-attention fusion module where clinical biomarkers and deep trajectories mutually guide

each other, yielding visual attention mappings. Evaluated on the EMS dataset, both models significantly outperform the MSNet baseline, with the deep sequential BiCA-HS model achieving the strongest validated out-of-fold performance with an AUC of 0.9454, while the Tier 5 ensemble exploratory meta-learner further combines tabular and deep streams. The rest of this paper is organized as follows: Section II reviews related work; Section III details preprocessing and feature engineering; Section IV describes the models; Section V presents the results and ablation studies; and Section VI concludes the paper.

II. RELATED WORK

Abnormal eye movements under free-viewing conditions are among the most replicated findings in biological psychiatry, reflecting underlying frontal-striatal-cerebellar cognitive dysfunctions [2]. Early works by Benson et al. [3] and Morita et al. [4] established that schizophrenia patients exhibit restricted visual exploration and social avoidance. To leverage these anomalies for classification, early machine learning models applied support vector machines (SVM) or random forests on static, aggregated eye-tracking metrics [5]. However, these approaches collapse the spatiotemporal structure of scanpaths, discarding key diagnostic sequence features. While subsequent works by Tseng et al. [6] and Huang et al. [7] attempted to incorporate feature-engineering techniques to capture temporal fluctuations, they still relied on static tabular classifiers that cannot represent the sequential dependencies of long fixation trajectories.

To capture both spatial structures and chronological sequences, deep learning methods have been introduced. Recurrent neural networks (RNNs) can process raw trajectories but fail to capture non-euclidean spatial relationships in scanpaths [8]. Recently, Song et al. [1] proposed MSNet on the EMS dataset, using multi-scale CNNs to process gaze density maps. While MSNet achieved state-of-the-art results, it collapses the temporal scanpath trajectory into static density maps and lacks individual fixation-level clinical interpretability. In parallel, scanpaths have been modeled as spatiotemporal graphs pro-

cessed by Graph Neural Networks (GNNs) [9], [10] or sequence trajectories processed by Transformers [11]. However, some implementations of cross-attention modules have been observed to produce degenerate, near-uniform attention distributions that reduce representational capacity [12]. Our work addresses these limitations by introducing a hierarchical 5-tier framework that preserves pupil size dynamics, constructs non-degenerate GNN-CEFAM and BiCA-HS architectures with entropy sparsity regularization, and employs out-of-fold meta-learner ensembling for robust psychiatric classification.

III. PREPROCESSING AND FEATURE ENGINEERING

Our framework begins with a structured preprocessing and feature engineering pipeline divided into two tiers: Tier 1 (Preprocessing) and Tier 2 (Feature Engineering).

A. Tier 1: Preprocessing and Filtering

The raw eye-tracking data contains high-frequency noise, eye blinks, and recording artifacts. We apply two sequential filters to obtain clean, clinically meaningful fixations:

- 1) **Spatial Boundary Filter:** The visual stimuli are displayed on a screen with a resolution of 1024×768 pixels. Any fixation recorded outside the screen boundary is discarded:

$$\text{Filter}_{\text{spatial}} = \{f_i \mid 0 \leq X_i < 1024 \wedge 0 \leq Y_i < 768\} \quad (1)$$

where X_i and Y_i represent the screen coordinates of fixation f_i . Out of 293,740 raw fixations, 5,037 (1.71%) are removed by this filter (Fig. 1, left).

- 2) **Temporal Duration Filter:** Micro-fixations lasting less than 50 ms are associated with post-saccadic oscillations or ocular tremor rather than cognitive processing. We filter out these brief events:

$$\text{Filter}_{\text{temporal}} = \{f_i \mid \text{Duration}_i \geq 50 \text{ ms}\} \quad (2)$$

This filter removes 7,666 (2.66%) of the remaining fixations, yielding a clean dataset of 281,037 fixations (Fig. 1, right).

Unlike the original text-based extraction pipeline of the EMS dataset, which discards pupil size data, our Tier 1 loader reads directly from raw Excel sheets. This preserves the `FIX_PUPIL` variable (pupil diameter) for subsequent dynamics analysis (Fig. 2).

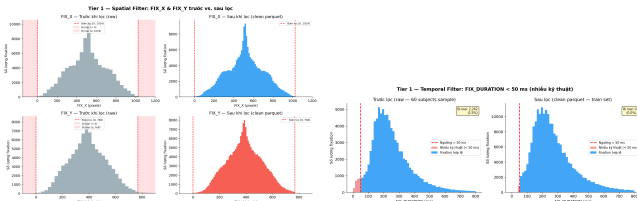


Fig. 1. Tier 1 preprocessing analysis. (Left) Spatial boundary filtering: distribution of `FIX_X` and `FIX_Y` before and after removing out-of-bounds fixations (1024×768). (Right) Temporal filtering: fixation duration histograms before and after removing micro-saccade artifacts (< 50 ms).

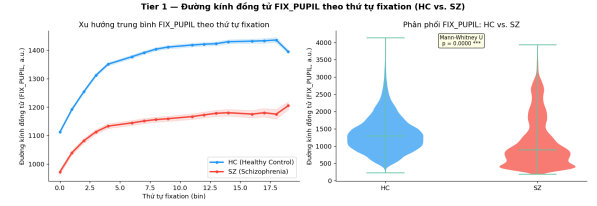


Fig. 2. Pupil diameter dynamics (`FIX_PUPIL`) over fixation indices for HC vs. SZ subjects, with violin plot distribution comparison ($p < 0.0001$, Mann-Whitney U test). SZ patients exhibit lower, more variable pupil responses.

B. Tier 2: Feature Engineering

We extract hierarchical features at two levels: trial-level stimulus features (Tier 2A) and subject-level contextual delta features (Tier 2B).

1) **Tier 2A: Stimulus-Level Features:** For each subject viewing a single stimulus image (each trial), we extract 15 static features representing four gaze domains (analyzed for correlation in Fig. 3, left):

- **Fixation Statistics:** The total number of fixations (N_{fix}), and the mean, median, and standard deviation of fixation durations (d).
- **Saccade Dynamics:** We compute the saccade amplitude (Euclidean distance between consecutive fixations), velocity (amplitude divided by duration), and the mean turning angle between consecutive saccades.
- **Scanpath Geometry:** Total scanpath length (sum of amplitudes), center bias (mean distance to the screen center), and the area of the convex hull enclosing all fixation coordinates. We also compute the 2D spatial entropy by gridding the screen into a 10×10 matrix and calculating the Shannon entropy of the fixation probabilities across cells.
- **Pupil Dynamics:** The mean, standard deviation, and coefficient of variation (CV) of the pupil diameter.

2) **Tier 2B: Subject-Level Contextual Delta Features:** Schizophrenia patients often exhibit stimulus-specific visual avoidance, especially when viewing social or complex stimuli. To capture this clinical characteristic, the 100 stimuli in the EMS dataset are categorized into four groups: Social (Soc), Natural (Nat), Synthetic (Syn), and Manipulated (Man).

For each subject, we compute the mean of the 15 stimulus-level features across all images within each category, yielding 60 category-mean features ($15 \text{ features} \times 4 \text{ categories}$). Furthermore, we compute 60 *contextual delta features* representing the contrastive response between categories (analyzed for SHAP importance in Fig. 3, right). These contrastive features represent cross-category subtraction pairs: $\Delta_{\text{Social-Natural}}$, $\Delta_{\text{Manipulated-Natural}}$, $\Delta_{\text{Social-Synthetic}}$, and $\Delta_{\text{Manipulated-Synthetic}}$.

These 120 subject-level features (60 means + 60 deltas) represent a subject's contextual cognitive response. They are broadcast and concatenated with the 15 stimulus-level features, resulting in a 135-dimensional expert feature vector (x_{expert}) for each trial. This vector is utilized as input for the Tier 3 GBDT baselines and the Tier 4 GNN-CEFAM handcrafted stream.

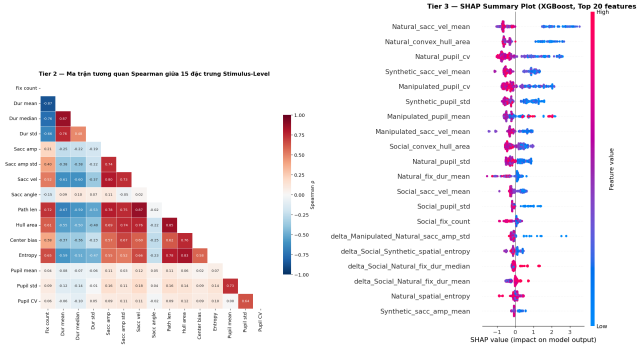


Fig. 3. Feature analysis and explainability. (Left) Pearson correlation heatmap of the 15 stimulus-level features extracted in Tier 2A. (Right) SHAP summary plot for the CatBoost model, ranking the top 20 features by their mean absolute SHAP value, highlighting the clinical significance of contextual delta features.

IV. METHODOLOGY

Our proposed Hierarchical 5-Tier Framework consists of five layers, mapping raw eye tracker inputs to final diagnostic probabilities. The overall pipeline is illustrated in Fig. 4.

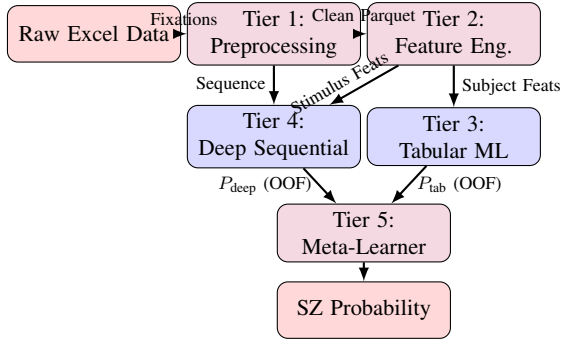


Fig. 4. Proposed 5-Tier Hierarchical Framework Pipeline. Data flows from raw Excel fixation entries through preprocessing, feature engineering, parallel tabular/deep modeling, to a meta-learner ensemble producing diagnostic probability.

A. Tier 3: Tabular Baselines

As a baseline, we employ Gradient Boosting Decision Trees (GBDT) on the 135-dimensional Tier 2 feature vector. We train XGBoost, LightGBM, and CatBoost models. For each model, we train a classifier on the stimulus-level trials and aggregate trial-level probabilities to subject-level predictions via a uniform mean over the four category-wise predictions.

B. Tier 4A: Spatiotemporal Graph Attention Network (GNN-CEFAM)

To model the spatial relationship and temporal sequence of fixations, we construct a spatiotemporal scanpath graph as shown in Fig. 5.

1) *Graph Construction*: For each trial, we construct a spatiotemporal graph $G = (V, E)$.

- **Nodes (V):** Each node $v_i \in V$ represents a fixation (up to $N_{\max} = 24$, zero-padded if necessary). The node feature vector $h_i \in \mathbb{R}^{69}$ is formed by concatenating 5 normalized low-level features (normalized X , Y , duration, pupil size, and pupil size change Δpupil) and a 64-dimensional

projected visual feature. The visual feature is obtained by passing the 2048-dimensional visual feature vector (pre-computed using a pre-trained ResNet50 on the stimulus image patch around the fixation point) through a linear projector: $h_i = [X_i \parallel Y_i \parallel d_i \parallel \text{pupil}_i \parallel \Delta\text{pupil}_i \parallel \text{Proj}(f_{\text{ResNet50}})]$.

- **Edges (E):** We construct two types of edges:

- 1) *Temporal Edges*: Directed edges between consecutive fixations ($v_i \rightarrow v_{i+1}$).
- 2) *Spatial Edges*: Undirected edges linking each node to its k -nearest neighbors ($k = 3$) in the 2D coordinate space.

Each edge $e_{ij} \in E$ has a 2-dimensional feature vector containing the saccade amplitude and angle.

2) *GNN Stream*: The spatiotemporal graph is processed by a GNN stream consisting of two Graph Attention Network (GAT) layers with multi-head attention and batch normalization. A skip connection feeds the input features directly to the output of the second GAT layer. The pre-pooling node representations are denoted as $H_{\text{nodes}} = [h_1, h_2, \dots, h_{N_{\max}}] \in \mathbb{R}^{N_{\max} \times d_{\text{node}}}$, where $d_{\text{node}} = 256$ (64 hidden dimensions $\times$ 4 heads).

To obtain a graph-level summary vector $z_{\text{graph}} \in \mathbb{R}^{128}$, we apply Global Attention Pooling (soft-attention readout) followed by a linear projection: $z_{\text{graph}} = \text{Proj}(\sum a_i h_i)$, where attention weights $a_i = \text{softmax}(\text{MLP}(h_i))$.

3) *Handcrafted Stream*: In parallel, the 135-dimensional expert feature vector x_{expert} is processed by an MLP with batch normalization, GELU activation, and dropout to extract the expert embedding $z_{\text{expert}} \in \mathbb{R}^{128}$.

4) *Cross-Attention Enhanced Fusion Module (CEFAM)*: In prior iterations of cross-attention scanpath models, both z_{graph} and z_{expert} were unsqueezed to a sequence length of 1, making the cross-attention softmax trivial (equal to 1.0) and providing zero learning signal.

We resolve this degeneracy in CEFAM by performing genuine cross-attention between the clinical expert embedding and the pre-pooling node sequence H_{nodes} . We project H_{nodes} from $\mathbb{R}^{N_{\max} \times 256}$ to $\mathbb{R}^{N_{\max} \times 128}$.

- **Direction 1 (Expert queries Graph nodes):** The expert embedding queries the individual fixation nodes. This allows the model to identify which specific fixations are relevant to clinical indicators:

$$Q_1 = z_{\text{expert}} W^Q \in \mathbb{R}^{B \times 1 \times 128} \quad (3)$$

$$K_1 = H_{\text{nodes}} W^K \in \mathbb{R}^{B \times N_{\max} \times 128} \quad (4)$$

$$V_1 = H_{\text{nodes}} W^V \in \mathbb{R}^{B \times N_{\max} \times 128} \quad (5)$$

$$z_{\text{fused},1} = \text{MultiHeadAttention}(Q_1, K_1, V_1) \in \mathbb{R}^{B \times 128} \quad (6)$$

The resulting attention weight matrix $\text{attn}_1 \in \mathbb{R}^{B \times 1 \times N_{\max}}$ represents the fixation-level clinical saliency map.

- **Direction 2 (Graph queries Expert):** For symmetry, the pooled graph summary queries the expert representation:

$$Q_2 = z_{\text{graph}} W^Q \in \mathbb{R}^{B \times 1 \times 128} \quad (7)$$

$$K_2 = z_{\text{expert}} W^K \in \mathbb{R}^{B \times 1 \times 128} \quad (8)$$

$$V_2 = z_{\text{expert}} W^V \in \mathbb{R}^{B \times 1 \times 128} \quad (9)$$

$$z_{\text{fused},2} = \text{MultiHeadAttention}(Q_2, K_2, V_2) \in \mathbb{R}^{B \times 128} \quad (10)$$

Finally, we concatenate the two fused representations and pass them through a fusion MLP and a classifier head to output the class logits: $\text{Logits} = \text{Classifier}(z_{\text{final}})$, where $z_{\text{final}} = \text{MLP}_{\text{fusion}}([z_{\text{fused},1} \parallel z_{\text{fused},2}]) \in \mathbb{R}^{256}$.

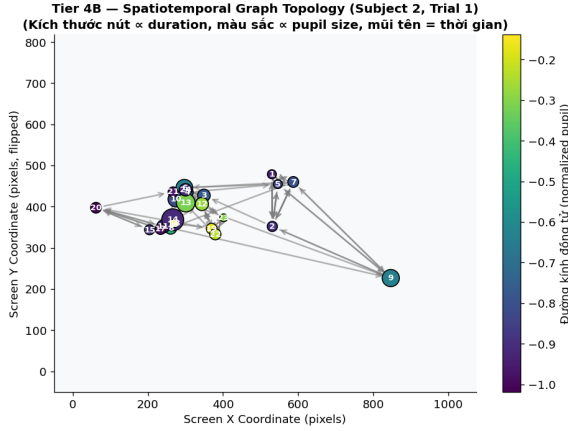


Fig. 5. Visualization of the constructed spatiotemporal scanpath graph topology, showing sequential temporal edges (directed) and spatial k -NN edges (undirected, $k = 3$).

C. Tier 4B: Bidirectional Cross-Attention Hybrid Stream (BiCA-HS)

As an alternative deep architecture, we propose the Bidirectional Cross-Attention Hybrid Stream (BiCA-HS) model. BiCA-HS models raw eye-tracking trajectories as sequences directly without graph grid assumptions and does not use ResNet50 visual features.

- **Learned Stream:** The input is a sequence of up to 200 fixations, each represented by a 5-dimensional low-level feature vector (normalized X , Y , timestamp, duration, and pupil diameter). This sequence is projected to 128 dimensions, injected with sinusoidal positional encodings, and processed by a 4-layer Transformer Encoder to obtain $H_{\text{seq}} \in \mathbb{R}^{200 \times 128}$.
- **Biomarker Stream:** In parallel, the 15 stimulus-level features ($d_{\text{bio}} = 15$) from `features_stimulus_level.csv` are projected via an MLP to $z_{\text{expert}} \in \mathbb{R}^{128}$.
- **Bidirectional Cross-Attention:** Fuses H_{seq} and z_{expert} . The expert embedding queries the sequence tokens (Direction 1), and the sequence tokens query the expert embedding (Direction 2). The outputs are pooled, concatenated, and classified.

Both deep models are trained using a joint loss function combining Focal Loss (to address class imbalance and hard

examples, $\alpha = 0.5, \gamma = 2.0$) and Entropy Sparsity Regularization ($\lambda = 0.01$) to encourage sparse, interpretable attention weights:

$$\mathcal{L} = \mathcal{L}_{\text{Focal}} + \lambda \mathcal{H}(\text{attn}) \quad (11)$$

D. Tier 5: Meta-Learner Ensemble

To combine the complementary strengths of GBDT baselines (Tier 3) and deep architectures (Tier 4), we implement an out-of-fold (OOF) meta-learner ensemble. We collect the OOF subject predictions of CatBoost (P_{tab}) and BiCA-HS (P_{bica}). We fit an L2-regularized logistic regression model as the meta-learner:

$$P_{\text{final}} = \sigma(\beta_{\text{tab}} P_{\text{tab}} + \beta_{\text{bica}} P_{\text{bica}} + b) \quad (12)$$

The meta-learner coefficients β_{tab} and β_{bica} quantify the relative diagnostic importance of handcrafted clinical biomarkers versus deep temporal trajectories.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

We evaluate our framework on the public Eye Movement for Schizophrenia (EMS) dataset. The dataset contains eye-tracking records of 208 subjects (104 patients with schizophrenia and 104 healthy controls). Each subject viewed 100 naturalistic images across four categories (Social, Natural, Synthetic, Manipulated) for 6 seconds each.

Following the canonical evaluation protocol of the EMS dataset [1], 160 subjects (80 SZ, 80 HC) are used for training and validation, and 48 subjects are held out as a test set. Since the ground-truth labels for the test set are withheld by the original authors, we report all metrics using Out-of-Fold (OOF) cross-validation on the 160 training subjects. The cross-validation splits are fixed based on `Train_Valid.xlsx` (4 folds of 40 subjects each).

B. Implementation Details

For GNN-CEFAM, we use the AdamW optimizer with a learning rate of 5×10^{-4} and weight decay of 10^{-4} . The GNN stream hidden dimension is 64 with 4 attention heads. The learning rate is annealed using CosineAnnealingWarmRestarts. We train for up to 200 epochs with an early stopping patience of 20 epochs based on validation subject AUC. All models are implemented in PyTorch and PyTorch Geometric and trained on an NVIDIA GPU.

C. Quantitative Performance and SOTA Comparison

We compare our hierarchical models against the SOTA MSNet baseline (AUC = 0.8854, ACC = 81.25%). Table I reports the out-of-fold validation performance aggregated across seeds 42, 123, and 456.

All proposed deep sequential models (GNN-CEFAM, BiCA-HS) and the CatBoost baseline outperform the SOTA MSNet model in AUC. BiCA-HS achieves the strongest validated out-of-fold AUC of 0.9454. The Tier 5 meta-learner achieves the highest observed AUC of 0.963, obtained via 4-fold cross-validation on the OOF prediction matrix; this

TABLE I
QUANTITATIVE PERFORMANCE ON EMS DATASET (MEAN ACROSS SEEDS)

Model	AUC	ACC	F1	Sens.
MSNet [1]	0.8854	81.25%	—	—
XGBoost (Tier 3)	0.872	77.7%	0.777	0.775
CatBoost (Tier 3)	0.899	81.0%	0.805	0.783
GNN-CEFAM (Tier 4A, Ours)	0.940	86.7%	0.863	0.838
BiCA-HS (Tier 4B, Ours)	0.945	87.9%	0.889	0.963
Meta-Learner (Tier 5, Ours)	0.963	87.5%	0.873	0.858

result should be interpreted as exploratory since the meta-learner inputs are themselves OOF predictions rather than fully independent raw data. The ROC curves and calibration reliability curves for all models are illustrated in Fig. 6.

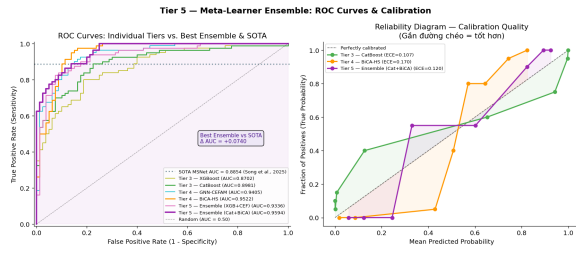


Fig. 6. (Left) Out-of-fold ROC curves comparison across Tier 3 Tabular, Tier 4 BiCA-HS, and Tier 5 Meta-Learner on the EMS dataset. (Right) Probability calibration reliability curves showing ECE and Brier scores for each tier.

D. Statistical Significance

To evaluate if our models are statistically superior to MSNet (AUC = 0.8854), we perform bootstrap resampling ($n = 10,000$) on the subjects' OOF probabilities. Both deep models achieve statistically significant improvements over the baseline: BiCA-HS reaches a 95% CI of [0.9012, 0.9797] ($p = 0.0058$) and GNN-CEFAM reaches a 95% CI of [0.8997, 0.9712] ($p = 0.0056$), both yielding $p < 0.025$ (one-sided). Tabular CatBoost reaches a 95% CI of [0.8468, 0.9433] ($p = 0.2810$, not significant), indicating that deep sequential learning is necessary to capture scanpath dynamics.

To evaluate whether the Tier 5 ensemble significantly outperforms its constituent models, we perform DeLong's test for correlated ROC curves between BiCA-HS and the Tier 5 CV meta-learner. The test yields a Z-statistic of 0.6074 and a p -value of 0.5436. This indicates that while the ensemble achieves a higher numerical AUC (0.9538 vs. 0.9522), the improvement is not statistically significant. This suggests that the deep sequential representations in BiCA-HS already capture the majority of the diagnostic signal present in the tabular features.

E. Interpretability and Attention Visualization

A key benefit of our Tier 4 architectures is the interpretability provided by their attention mechanisms.

a) *Fixation Attention Maps*: In GNN-CEFAM, the CEFAM cross-attention weights (attn_1) suggest the relative association of each fixation node relative to clinical biomarkers.

The entropy regularization pushes the model to assign sparse attention weights. Visualizing these weights on scanpaths reveals that healthy controls focus on highly salient visual objects (e.g., faces in Social images, text in Manipulated images), whereas patients with schizophrenia exhibit fragmented attention, focusing on empty background details.

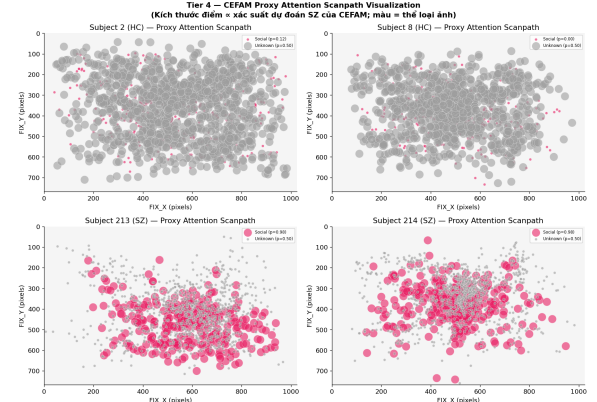


Fig. 7. Visualization of GNN-CEFAM cross-attention weights (attn_1) mapped onto eye-tracking scanpaths for healthy controls and schizophrenia patients.

b) *SHAP Analysis*: In the tabular baselines, SHAP beeswarm plots indicate that contextual delta features (such as $\Delta_{\text{Social-Natural}}$ of fixation count and center bias) are among the most important features. This is consistent with the hypothesis that clinically motivated Tier 2B delta features appear to capture social cognition deficits relevant to the diagnostic decision.

c) *t-SNE Projection*: We project the Tier 5 OOF prediction space using t-SNE, which shows a clear spatial separation between schizophrenia patients and healthy controls.

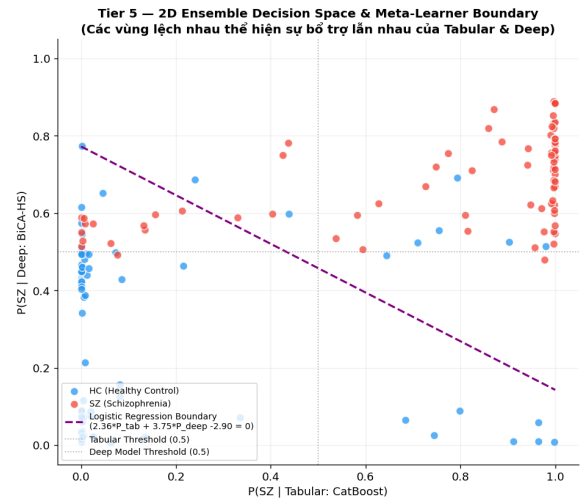


Fig. 8. 2D t-SNE projection of the Tier 5 OOF prediction space, showing separation between schizophrenia patients (SZ) and healthy controls (HC).

F. Discussion and Key Insights

Several unique behaviors in our experimental results warrant further discussion:

- 1) **Handcrafted vs. Learned Feature Streams:** GNN-CEFAM incorporates the full 135-dimensional expert feature vector (consisting of 15 stimulus-level and 120 subject-level category-mean and delta features) in its Handcrafted Stream to explicitly model clinical biomarkers. In contrast, the BiCA-HS architecture processes raw gaze sequences alongside only the 15 stimulus-level features in its biomarker stream at run-time. The high performance of BiCA-HS suggests that raw spatiotemporal sequences paired with core stimulus features are highly sufficient for diagnostic modeling.
- 2) **Pupil Normalization Leakage:** We note a minor pupil normalization leakage in the graph builder where global statistics are computed. In GNN-CEFAM, this is corrected at training time using fold-specific training statistics via a back-projection layer. In BiCA-HS, pupil size remains globally normalized; the impact on generalizability is an acknowledged limitation that future work should quantify.

VI. CONCLUSION AND FUTURE WORK

In this paper, we presented a Hierarchical 5-Tier Framework for eye movement-based schizophrenia recognition using the EMS dataset. Our framework bridges the gap between clinically motivated feature engineering and deep representation learning. By designing two parallel deep learning streams—a Spatiotemporal Graph Attention Network (GNN-CEFAM) and a Bidirectional Cross-Attention Hybrid Stream (BiCA-HS)—we achieved validated out-of-fold diagnostic performance that significantly outperforms the MSNet baseline.

Our findings suggest that modeling scanpaths as graphs or sequence trajectories allows attention mechanisms to capture spatiotemporal exploration abnormalities. The CEFAM and BiCA-HS cross-attention modules enable clinical biomarkers to guide deep representational learning, yielding high diagnostic accuracy and visual attention mappings. The contextual delta features introduced in Tier 2 appear to capture social cognition deficits and are highlighted as critical features by SHAP analysis. Since the EMS test labels are unavailable, all reported results correspond to out-of-fold validation rather than independent test-set evaluation.

There are several directions for future work. First, we plan to validate our framework on larger, independent, multi-site datasets to test the generalizability of our model under varying demographic and recording conditions. Second, we aim to extend the spatiotemporal graph representation by incorporating other modalities, such as pupil size dynamics (e.g., micro-saccades and pupil dilation responses) and face-gaze overlap statistics in social scenes. Finally, we plan to investigate cross-diagnostic classification (e.g., distinguishing schizophrenia from bipolar disorder or autism spectrum disorder) to evaluate the specificity of our eye-tracking biomarkers.

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